

Roman Kochnev

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Education

University of Würzburg , M.Sc. Computer Science (Informatik) • Focus: AutoML, Large Language Models, NLP • Thesis: Enabling Symbolic Reasoning with Advanced RAG Systems	2023 – 2026
Lomonosov Moscow State University , M.Sc. Applied Mathematics and Computer Science	2020 – 2022
National University of Science and Technology MISIS , B.Sc. Applied Mathematics	2016 – 2020

Experience

Data Science Researcher (Master's Thesis) , Bosch Center for AI – Renningen, Germany • Developed APOLO, casting ontology learning as prompt optimization: LLM-based ontology learners tuned with DSPy (GEPA) without weight updates outperform SOTA baselines; accepted at ISWC 2026 • Built GRAIL, a reasoning-focused QA benchmark for evaluating RAG and GraphRAG systems on OWL ontologies, with reasoner-verified ground truth across 1500+ questions • Designed OWL2Doc, a multi-agent pipeline (Planner, Writer, Critic) generating annotated text-ontology pairs from expert-curated OWL ontologies	Jan 2026 – Jul 2026
Research Assistant (HiWi) , University of Würzburg – Würzburg, Germany • First author of NNGPT (CVPRW 2026), an AutoML framework where an LLM generates PyTorch architectures, tunes hyperparameters and self-improves via RL feedback • Co-developed the original LEMUR neural-network dataset; co-author of LEMUR 2 (CVPRW 2026) • First author of Optuna vs Code Llama (ICCVW 2025), benchmarking LLM-based hyperparameter tuning against Optuna	Jun 2025 – Present
Software Developer and Analyst , ROSBANK, formerly Société Générale Group – Moscow • Built and maintained machine learning models for core banking use cases • Delivered analyses, reports and visualizations that informed business decisions • Sourced, cleaned and validated data from closed internal banking systems • Partnered with cross-functional teams to turn company data into business solutions; mentored interns and junior colleagues	May 2022 – Sep 2023
Data Analyst , Glowbyte Consulting – Moscow • Developed and optimized PL/SQL scripts, built Power BI dashboards and ran A/B tests for analytics projects across top-tier banks and major food retail chains	Sep 2021 – Apr 2022

Publications

1. **R. Kochnev**[†], H. T. Mai[†], C. X. Chu, L. Lange, H. Paulheim, D. Stepanova, "APOLO: Automatic Prompt Optimization for Ontology Learning", *ISWC*, 2026. ([†]equal contribution)
2. **R. Kochnev**, W. Khalid, T. Uzun, X. Zhang, Y. Dhameliya, F. Qin, C. Vysyaraju, R. Duvvuri, A. Goyal, D. Ignatov, R. Timofte, "NNGPT: Rethinking AutoML with Large Language Models", *CVPRW*, 2026.
3. T. Uzun, W. Khalid, S. Din, S. Mulukuledu, A. Singh, C. Vysyaraju, R. Duvvuri, A. Goyal, Y. R. Lukhi, M. A. Hussain, K. Jesani, U. Shrestha, Y. Mittal, **R. Kochnev**, P. Kadam, M. Ikram, H. Moradiya, A. Arslanian, D. Ignatov, R. Timofte, "LEMUR 2: Unlocking Neural Network Diversity for AI", *CVPRW*, 2026.
4. A. Goodarzi, **R. Kochnev**, W. Khalid, H. Goudarzi, F. Qin, T. Uzun, Y. Dhameliya, Y. Kathiriya, Z. Bentyn, D. Ignatov, R. Timofte, "LEMUR Neural Network Dataset: Towards Seamless AutoML", *arXiv*, 2025.
5. **R. Kochnev**, A. Goodarzi, Z. Bentyn, D. Ignatov, R. Timofte, "Optuna vs Code Llama: Are LLMs a New Paradigm for HPO?", *ICCVW*, 2025.
6. **R. Kochnev**, R. Senchenko, "Intelligent Method for Real Estate Price Prediction in Moscow Using a Machine Learning Model" (in Russian), *Proceedings of the 75th Days of Science of NUST MISIS Students*, 2020. ISBN 978-5-907227-11-8.

Technical Skills

Programming Languages: Python, SQL, Java, C++
ML / Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, PEFT/LoRA, NumPy, Pandas, OpenCV
LLM & AutoML: Prompt Optimization (DSPy, GEPA), RAG Pipelines, Ontology-based Reasoning, Neural Architecture Search
MLOps & Infra: Docker, Azure, MLflow, Kubernetes, Git
Tools: Jira, Confluence, Power BI, Excel, $\LaTeX$
Languages (spoken): English (C1), German (B1), Russian (Native)